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Assessing the Robustness of CNN and Hybrid Classifiers for Subject-Independent ADHD Detection from Raw EEG

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Abstract:	Attention-deficit/hyperactivity disorder (ADHD) is commonly studied using electroencephalogram (EEG) signals combined with convolutional neural network (CNN)-based feature extraction and classical machine learning classifiers. Prior studies report high classification accuracy using raw EEG inputs, typically evaluated on a single, fixed train-test split, raising concerns about generalizability to unseen subjects. We assess the subject-independent generalization of a CNN-based raw-EEG ADHD classification framework, coupling a two-stage spatial-temporal CNN with five classical hybrid classifiers (logistic regression, non-linear support vector machine, random forest, Gaussian naive Bayes, and -nearest neighbors), using grouped 10-fold cross validation on a publicly available pediatric ADHD/Control EEG dataset (121 subjects, 508 epochs). Pooled cross-validated accuracy across all six classifiers ranged from approximately 65% to 70%, substantially lower than single-split accuracies reported in prior literature. Paired Wilcoxon signed-rank tests across folds revealed no statistically significant difference between the CNN and any classical hybrid classifier (all > 0.05). A repeated random single-split analysis ($= 50$) showed that several previously reported accuracies fall within the range achievable through favorable sample splitting alone, while others exceed the maximum accuracy observed across all repeated splits, suggesting split selection alone does not fully explain prior results. We additionally evaluated CORAL-based domain adaptation to improve cross-subject generalization; this yielded no statistically significant improvement (≈ 0.30). These findings quantify subject-dependent performance inflation in raw-EEG ADHD classification and establish a subject-independent benchmark for future work.
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06.07.2026

To the Editor,

Computer Methods and Programs in Biomedicine Update

Subject: Submission of manuscript "Assessing the Robustness of CNN and Hybrid Classifiers for Subject-Independent ADHD Detection from Raw EEG"

Dear Editor,

I am pleased to submit my manuscript titled "Assessing the Robustness of CNN and Hybrid Classifiers for Subject-Independent ADHD Detection from Raw EEG" for consideration as an Original Research article in Computer Methods and Programs in Biomedicine Update.

This manuscript presents a subject-independent validation framework for raw-EEG, CNN-based ADHD classification, combining a two-stage spatial-temporal convolutional architecture with five classical hybrid classifiers, evaluated using grouped 10-fold cross-validation on a publicly available pediatric ADHD/Control EEG dataset. The study quantifies the gap between single-split and subject-independent cross-validated performance across six classifiers, applies formal statistical testing to classifier comparisons, characterizes this gap further through repeated random single-split resampling, and evaluates CORAL-based domain adaptation as a candidate method for improving cross-subject generalization.

This work directly aligns with the scope of Computer Methods and Programs in Biomedicine Update, contributing a rigorously validated computational methodology with direct relevance to the clinical translatability of EEG-based biomarkers for ADHD. The findings offer a subject-independent benchmark and a cautionary methodological reference point for the growing body of deep learning research applied to EEG-based neurodevelopmental disorder detection, an area of active interest to this journal's readership.

This manuscript has not been published previously and is not under consideration for publication elsewhere, in whole or in part. I have no conflicts of interest to declare. The dataset used in this study is publicly available, and this is stated in the Data Availability section of the manuscript.

Thank you for considering this manuscript for publication. I look forward to your response.

Sincerely,

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Keywords

- Electroencephalography
- ADHD
- Cross-validation
- Generalization
- Domain adaptation

Highlights

- CNN-based raw-EEG ADHD classifiers show inflated single-split accuracy.
- Grouped 10-fold cross-validation reduces accuracy to 65-70\% across models.
- No classifier significantly outperforms others under subject-independent CV.
- Repeated splits show some reported accuracies exceed all resampled outcomes.
- CORAL domain adaptation does not significantly improve generalization.

Assessing the Robustness of CNN and Hybrid Classifiers for Subject-Independent ADHD Detection from Raw EEG

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ABSTRACT

Attention-deficit/hyperactivity disorder (ADHD) is commonly studied using electroencephalogram (EEG) signals combined with convolutional neural network (CNN)-based feature extraction and classical machine learning classifiers. Prior studies report high classification accuracy using raw EEG inputs, typically evaluated on a single, fixed train–test split, raising concerns about generalizability to unseen subjects. We assess the subject-independent generalization of a CNN-based raw-EEG ADHD classification framework, coupling a two-stage spatial–temporal CNN with five classical hybrid classifiers (logistic regression, non-linear support vector machine, random forest, Gaussian naive Bayes, and k -nearest neighbors), using grouped 10-fold cross-validation on a publicly available pediatric ADHD/Control EEG dataset (121 subjects, 508 epochs). Pooled cross-validated accuracy across all six classifiers ranged from approximately 65% to 70%, substantially lower than single-split accuracies reported in prior literature. Paired Wilcoxon signed-rank tests across folds revealed no statistically significant difference between the CNN and any classical hybrid classifier (all $p > 0.05$). A repeated random single-split analysis ($n = 50$) showed that several previously reported accuracies fall within the range achievable through favorable sample splitting alone, while others exceed the maximum accuracy observed across all repeated splits, suggesting split selection alone does not fully explain prior results. We additionally evaluated CORAL-based domain adaptation to improve cross-subject generalization; this yielded no statistically significant improvement ($p \approx 0.30$). These findings quantify subject-dependent performance inflation in raw-EEG ADHD classification and establish a subject-independent benchmark for future work.

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is one of the most commonly diagnosed neurodevelopmental conditions in children, with prevalence estimates in the range of 5–7% worldwide [1]. Left unmanaged, ADHD is associated with long-term psychosocial, educational, and occupational difficulties, and its clinical presentation frequently persists into adulthood [2]. Diagnosis currently relies primarily on clinical behavioral assessment guided by criteria such as the DSM [3] or ICD [4]; this process is time-consuming, requires specialist involvement, and is subject to inter-rater variability, motivating substantial interest in electroencephalography (EEG) as a candidate objective biomarker for ADHD. EEG offers high temporal resolution, relatively low cost compared to neuroimaging alternatives, and practical suitability for use with children [5, 6].

A large body of prior work has explored EEG-based ADHD classification using hand-crafted features – including non-linear and entropy-based measures [7–9], connectivity metrics [10, 11], and wavelet or spectral analyses [12, 13] – typically paired with classical machine learning classifiers. More recently, deep learning approaches have been applied directly to raw or minimally processed EEG signals, removing the need for manual feature engineering [14–17]. This mirrors a broader trend in EEG-based decoding generally: Schirrneister et al. [18] showed that deep convolutional networks trained directly on raw EEG could match or exceed the performance of the widely used filter bank common spatial patterns (FBCSP) algorithm for motor-related decoding tasks. Among ADHD-specific studies, TaghiBeyglou et al. [19] proposed a convolutional neural network (CNN) architecture combining spatial and temporal convolutional blocks with raw EEG input, further hybridized with classical classifiers (logistic regression, support vector machines, random forests, Gaussian naive Bayes, and k -nearest neighbors) trained on CNN-derived features, reporting accuracies as high as 95.83% on a publicly available pediatric ADHD/Control EEG dataset [20].

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Whether deep learning consistently outperforms classical machine learning for EEG classification tasks, however, remains an open and task-dependent question. Several recent studies have found that carefully engineered features paired with classical classifiers can match or exceed deep learning performance, particularly under resource-constrained or noisy conditions; for example, Paredes Ocaranza et al. [21] recently reported that traditional machine learning with domain-specific feature engineering outperformed EEGNet, a widely used deep learning architecture, for consumer-grade EEG emotion recognition, especially under cross-dataset evaluation. This inconsistency across the literature underscores the importance of rigorous, generalizable evaluation protocols when comparing classifier families for any specific EEG classification task, including ADHD detection.

A common feature of prior EEG-based ADHD deep learning work, including [19], is that model performance is typically reported on a single, fixed train–test split of the data. This raises a methodological concern that has received growing attention across EEG-based deep learning research more broadly: when EEG recordings are segmented into short epochs and epochs (rather than subjects) are randomly assigned to training and test sets, data from the same subject can appear in both splits. This subject-level data leakage can allow a model to learn subject-specific idiosyncrasies rather than generalizable disease-related patterns, substantially inflating reported performance [22]. Given that a clinically useful EEG biomarker must generalize to new, previously unseen patients, evaluating performance under subject-independent conditions is essential for assessing real-world applicability.

In this study, we examine the extent to which reported classification performance for a raw-EEG, CNN-based ADHD detection framework holds under subject-independent evaluation. Using the same publicly available dataset [20], we implement a CNN architecture following the design principles described by TaghiBeyglou et al. [19] and evaluate it, together with five classical hybrid classifiers, using grouped k -fold cross-validation in which all epochs from a given subject are assigned exclusively to either the training or the test set. We further investigate whether CORAL-based unsupervised domain adaptation [23, 24] – a technique that has shown benefit for cross-subject and cross-dataset generalization in other EEG classification tasks such as emotion recognition [21] – can improve subject-independent performance in this setting. The contributions of this work are as follows:

1. We quantify the gap between single-split and subject-independent cross-validated performance across six classifiers (a CNN and five classical hybrid classifiers), with formal paired statistical testing of differences across folds.
2. We characterize this gap further through repeated random single-split resampling, examining whether previously reported accuracies fall within the range achievable through favorable sample splitting alone, or whether they exceed what such resampling can produce.
3. We evaluate CORAL-based unsupervised domain adaptation as a candidate method for improving cross-subject generalization in this specific task, an intervention not previously tested in this context to our knowledge.

2. Related Work

2.1. EEG-based ADHD classification

Early work on EEG-based ADHD detection relied heavily on manually engineered features. Ghassemi et al. [7] and Khoshnoud et al. [8] explored non-linear EEG features under resting and task conditions, while Mohammadi et al. [25] and Allahverdy et al. [26] combined non-linear features with shallow neural network classifiers. Connectivity-based approaches, such as phase transfer entropy analysis [11] and functional connectivity comparisons [10], have also been proposed as a means of capturing network-level disruptions associated with ADHD.

More recent studies have applied deep learning directly to EEG data. Chen et al. [14, 15] proposed deep learning frameworks using EEG-based brain networks and spatial-frequency representations, while Dubreuil-Vall et al. [16] used event-related spectral perturbations with a CNN to discriminate adult ADHD from healthy controls. TaghiBeyglou et al. [19] extended this line of work by proposing a CNN operating directly on raw EEG signals, avoiding time–frequency transformation altogether, and additionally used the CNN’s learned feature representations to train classical hybrid classifiers, reporting improved performance over the CNN alone on a single held-out test set. Our study uses the same publicly available dataset [20] that underlies this and several related studies from the same research group [11, 25, 26].

A recent preprint by Amini et al. [27] proposed ADHDeepNet, a raw-EEG deep learning model incorporating attention mechanisms and explainability techniques, evaluated on the same 121-subject dataset used in the present study. Notably, that work employed a nested 10-fold cross-subject validation strategy rather than a single fixed split,

reflecting a similar methodological concern to the one motivating the present study. Our work differs in focus: rather than proposing a new architecture, we specifically quantify how much previously reported performance depends on validation methodology, and additionally test whether domain adaptation can improve cross-subject generalization for an existing, previously published architecture.

2.2. Classical machine learning versus deep learning for EEG classification

The relative merits of classical machine learning and deep learning for EEG-based classification remain task-dependent and actively debated. Schirrmester et al. [18] demonstrated that deep convolutional networks trained end-to-end on raw EEG could match or slightly exceed the performance of the FBCSP algorithm, a strong classical baseline, for motor decoding tasks. In contrast, more recent work by Paredes Ocaranza et al. [21] found that traditional machine learning with domain-specific feature engineering outperformed EEGNet, a compact deep learning architecture, for consumer-grade EEG emotion recognition, with the performance gap widening under cross-dataset evaluation – underscoring that architectural complexity does not universally translate to better generalization, particularly under distribution shift. This inconsistency across tasks and datasets motivates careful, task-specific evaluation of classifier comparisons, which is one focus of the present study.

2.3. Data leakage and validation practices in EEG deep learning

Concerns about subject-level data leakage in EEG deep learning studies have been raised across multiple clinical domains. Brookshire et al. [22] surveyed the translational EEG deep learning literature and found that a majority of studies use segment-based (rather than subject-based) holdout strategies, and demonstrated empirically, across Alzheimer’s disease and epilepsy classification tasks, that this can lead to substantially inflated reported performance relative to subject-independent evaluation. That survey explicitly includes [19] among the translational DNN-EEG studies examined. The present study extends this line of methodological inquiry with a focused, quantitative re-evaluation of one specific published framework, including repeated resampling analysis and a domain adaptation intervention not considered in prior leakage-focused work.

2.4. Domain adaptation for cross-subject generalization

Domain adaptation methods have been widely applied to address cross-subject and cross-dataset variability in EEG-based classification tasks such as emotion recognition and motor imagery decoding. Correlation alignment (CORAL) [23] and its deep learning extension, Deep CORAL [24], align the second-order statistics (covariance structure) of source and target feature distributions without requiring labeled target-domain data. Paredes Ocaranza et al. [21] used CORAL as part of a broader domain adaptation pipeline for cross-dataset EEG emotion recognition, reporting it as a useful component alongside subspace-learning methods. In the context of ADHD specifically, Mengi and Malhotra [28] proposed an unsupervised multisource domain adaptation framework – evaluating CORAL among other feature-alignment baselines – for harmonizing multi-site structural and functional MRI data in the ADHD-200 dataset. To our knowledge, domain adaptation techniques such as CORAL have not previously been evaluated for cross-subject generalization in raw-EEG, CNN-based ADHD classification specifically, which we address in this study.

3. Methods

3.1. Dataset

We used a publicly available EEG dataset comprising recordings from 61 children diagnosed with ADHD and 60 healthy control children, aged 7 to 12 years, originally collected by Nasrabadi et al. and released via IEEE DataPort [20]. EEG was recorded from 19 electrodes following the 10-20 system (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Cz, Pz) during a visual attention task in which children counted cartoon characters displayed on screen.

Recording length varied across subjects (12,258 to 43,252 samples per subject in our downloaded copy of the dataset). Based on the description of the original dataset in [19], this is presumed to correspond to a sampling rate of 128 Hz; however, no sampling-rate metadata was present in the released files themselves, and this should therefore be treated as an assumption rather than a confirmed fact. Readers attempting to reproduce this work are encouraged to verify this independently or contact the original data providers. This is a different, non-epoched version of the dataset than the fixed 512 Hz, 30-second-epoch version used by TaghiBeyglou et al. [19], which was prepared separately for a national EEG processing competition.

3.2. Preprocessing and epoching

Each subject's continuous recording was segmented into non-overlapping 30-second epochs (3,840 samples at the assumed 128 Hz sampling rate), discarding any remaining partial segment shorter than 30 seconds. This yielded 508 epochs in total (288 ADHD, 220 Control) across 121 subjects. Each epoch was normalized independently, per channel, using z-score standardization (subtracting the channel mean and dividing by the channel standard deviation within that epoch), following standard practice for deep learning on raw EEG. Figure 1 shows representative single-subject epochs (one ADHD, one Control) before and after this normalization step; these are illustrative single examples of the preprocessing pipeline and should not be interpreted as group-level or diagnostic patterns.

No additional preprocessing – such as bandpass filtering, artifact rejection, or re-referencing – was applied, consistent with the raw-EEG, transformation-free approach described in [19].

Representative Raw EEG Epochs Before and After Normalization
(fs=128Hz assumption; single representative examples, not group averages)

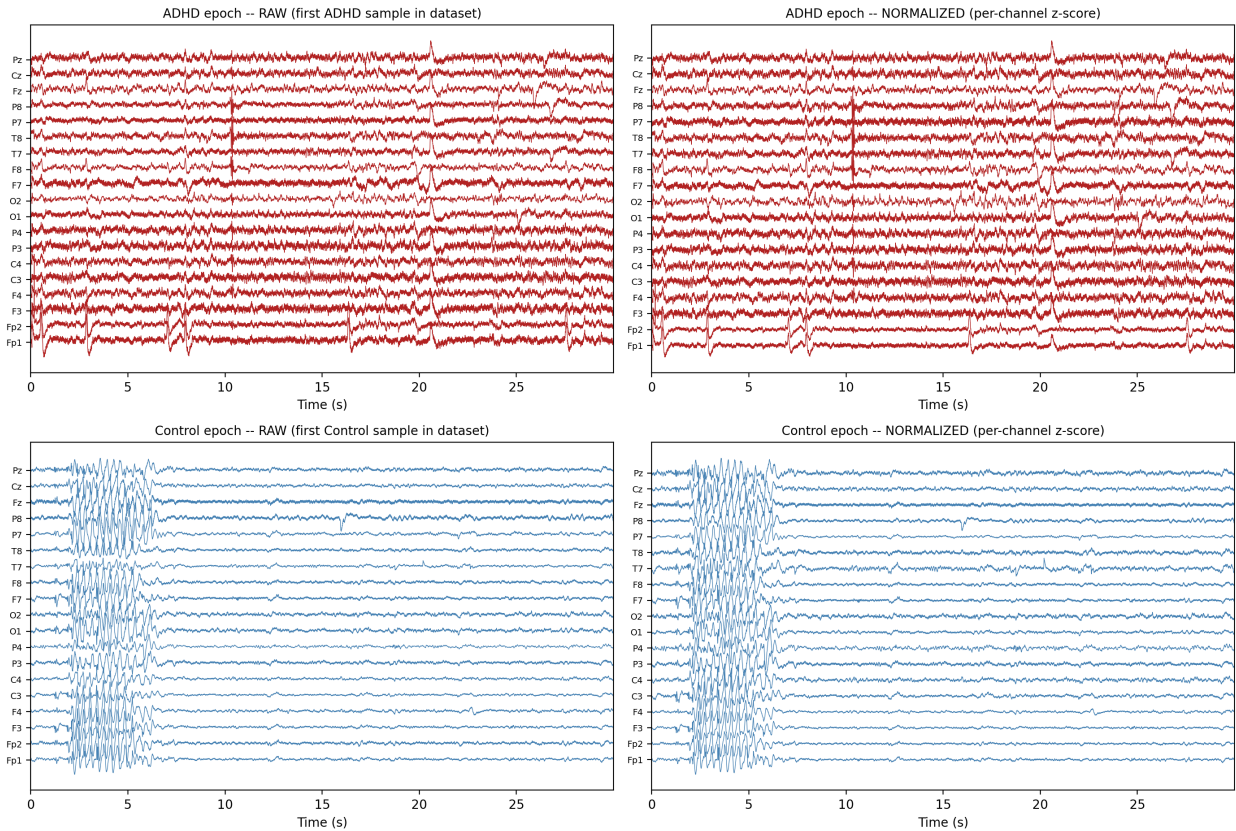


Figure 1: Representative raw and normalized EEG epochs for one ADHD subject and one Control subject (single examples, not group averages).

3.3. CNN architecture

We implemented a convolutional neural network following the two-stage spatial-temporal design described by TaghiBeyglou et al. [19], adapted to our input dimensions (19 channels $\times$ 3,840 time samples). The architecture consists of two spatial convolutional blocks (Conv2D with kernel sizes 10×1 and 4×1 , respectively, each followed by batch normalization and average pooling), which progressively reduce the channel dimension from 19 to 1; two temporal convolutional blocks (Conv1D with kernel sizes scaled to the sampling rate, each followed by batch normalization and average pooling), operating on the time axis after the channel dimension has been reshaped away; and a feature selection block consisting of a flatten operation followed by two dense layers (64 and 32 units). A final sigmoid-activated single neuron performs binary classification (ADHD vs. Control). The complete architecture is shown in Figure 2, and the

Table 1

Verified layer properties of the implemented CNN architecture (19 channels, 3840 time samples, assumed 128 Hz).

Layer	Filters/Units	Kernel size	Output shape
Input	—	—	$19 \times 3840 \times 1$
Spatial Conv 1	16	10×1	$10 \times 3840 \times 16$
Batch Norm	—	—	$10 \times 3840 \times 16$
AvgPool	—	2×1	$5 \times 3840 \times 16$
Spatial Conv 2	16	4×1	$2 \times 3840 \times 16$
Batch Norm	—	—	$2 \times 3840 \times 16$
AvgPool	—	2×1	$1 \times 3840 \times 16$
Reshape (squeeze channel)	—	—	3840×16
Temporal Conv 1	32	32	3809×32
Batch Norm	—	—	3809×32
AvgPool	—	16	238×32
Temporal Conv 2	32	16	223×32
Batch Norm	—	—	223×32
AvgPool	—	8	27×32
Flatten	—	—	864
Dense 1	64	—	64
Dense 2	32	—	32
Classification	1	—	1

corresponding verified layer-by-layer tensor dimensions are listed in Table 1. All intermediate tensor shapes were confirmed by direct inspection of the implemented model rather than assumed from the architectural description alone.

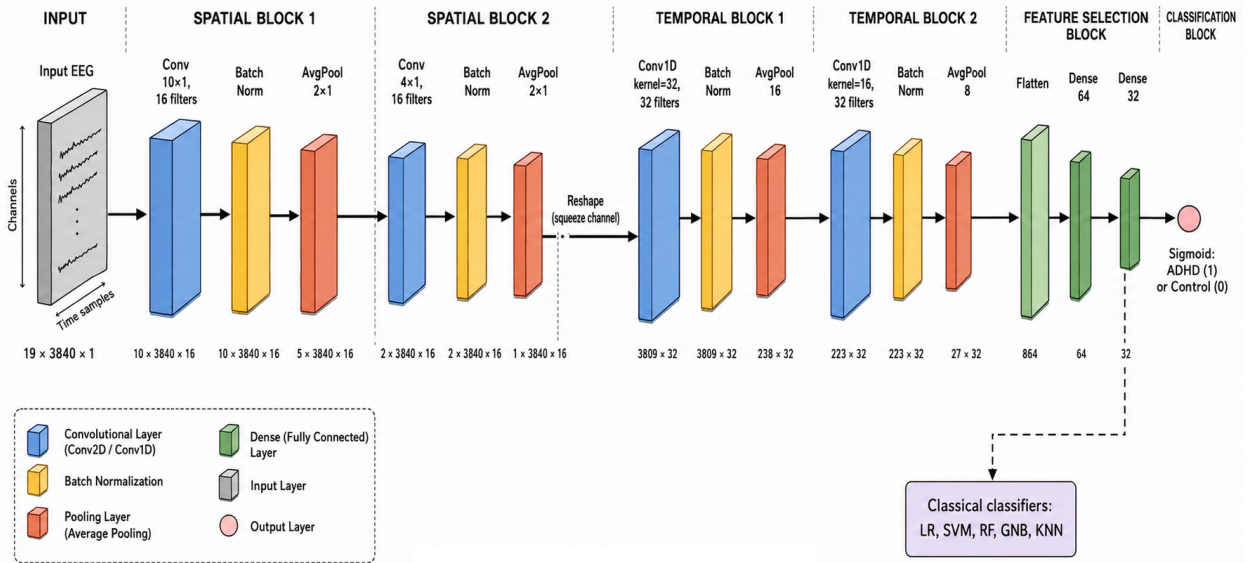


Figure 2: CNN architecture used in this study, showing the two spatial convolutional blocks, two temporal convolutional blocks, feature selection block, and classification block. Dashed arrow indicates that Dense-layer features are additionally extracted to train the classical hybrid classifiers described in Section 3.4.

Table 2

Classical classifier configurations.

Classifier	Key parameters	Class weighting
Logistic Regression (LR)	L2 penalty, max_iter=2000	Balanced
Non-linear SVM (NLSVM)	RBF kernel	Balanced
Random Forest (RF)	100 trees	Balanced
Gaussian Naive Bayes (GNB)	Default	None (unsupported)
k -Nearest Neighbors (KNN)	$k = 5$	None (unsupported)

3.4. Hybrid classical classifiers

Following [19], we extracted the CNN's learned feature representations at the output of the first dense layer (64-dimensional) and used them to train five classical classifiers: logistic regression (LR) with L2 regularization, a non-linear support vector machine (NLSVM) with a radial basis function kernel, a random forest (RF) with 100 trees, Gaussian naive Bayes (GNB), and k -nearest neighbors (KNN, $k = 5$). Given class imbalance in the dataset (57% ADHD, 43% Control), LR, NLSVM, and RF were trained with balanced class weighting; GNB and KNN do not support class weighting in their standard scikit-learn implementations and were trained with default settings. Table 2 summarizes the configuration of each classifier.

3.5. Evaluation metrics

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives respectively, with ADHD treated as the positive class. We report:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision}_{\text{ADHD}} = \frac{TP}{TP + FP}, \quad \text{Precision}_{\text{Control}} = \frac{TN}{TN + FN} \quad (2)$$

$$\text{Sensitivity} = \text{Recall}_{\text{ADHD}} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \text{Recall}_{\text{Control}} = \frac{TN}{TN + FP} \quad (3)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

AUC denotes the area under the receiver operating characteristic (ROC) curve.

3.6. Validation strategy

To evaluate subject-independent generalization, we used grouped 10-fold cross-validation (via scikit-learn's GroupKFold), ensuring that all epochs belonging to a given subject were assigned exclusively to either the training or the test fold, preventing subject-level data leakage [22]. The CNN was retrained independently for each fold (100 maximum epochs, batch size 16, Adam optimizer, learning rate 1×10^{-4} , early stopping on validation loss with patience 8), and classical classifiers were trained on that fold's extracted CNN features. Predictions were pooled across all 10 folds (508 total out-of-fold predictions) to compute overall accuracy, precision, recall, F1-score, and AUC for each classifier.

To test whether performance differences between the CNN and each classical hybrid classifier were statistically meaningful, we performed paired Wilcoxon signed-rank tests across the 10 folds, comparing per-fold accuracy (and separately, per-fold AUC) between the CNN and each of the five classical classifiers.

To further characterize how much of previously reported single-split performance might be attributable to favorable sample splitting alone, we additionally performed repeated random single-split evaluation ($n = 50$ independent 80/20 grouped splits, GroupShuffleSplit), training and evaluating all six classifiers on each split, and compared the resulting distribution of accuracies against previously reported single-split figures.

Table 3

Pooled performance metrics across 508 out-of-fold predictions (10 grouped folds).

Classifier	Accuracy (%)	Precision _{ADHD}	Precision _{Control}	Sensitivity	Specificity	F1	AUC
CNN	65.16	0.756	0.574	0.569	0.759	0.650	0.788
LR	66.34	0.702	0.612	0.705	0.609	0.704	0.750
NLSVM	67.72	0.709	0.632	0.729	0.609	0.719	0.751
RF	64.57	0.670	0.605	0.740	0.523	0.703	0.739
GNB	70.28	0.683	0.759	0.889	0.459	0.772	0.711
KNN	66.73	0.679	0.646	0.785	0.514	0.728	0.727

3.7. Domain adaptation

To test whether cross-subject generalization could be improved through domain adaptation, we applied CORAL [23], an unsupervised method that aligns the second-order statistics (covariance) of source and target feature distributions via a closed-form linear transformation, without requiring target-domain labels. The notation below reflects our own implementation of the method's stated principle, following Sun, Feng, and Saenko (2016) [23], rather than a direct transcription of the original paper's equations.

Given source (training-fold) features $D_S \in \mathbb{R}^{n_S \times d}$ and target (test-fold) features $D_T \in \mathbb{R}^{n_T \times d}$, let C_S and C_T denote their respective (regularized) feature covariance matrices. The source features are whitened using C_S and re-colored using C_T :

$$D_S^* = (D_S - \bar{D}_S) C_S^{-1/2} C_T^{1/2} + \bar{D}_T \quad (5)$$

where $\bar{D}_S$ and $\bar{D}_T$ denote the feature-wise means of the source and target sets, and $C_S^{-1/2}$, $C_T^{1/2}$ are computed via eigendecomposition. CORAL was applied to the 64-dimensional Dense-1 features prior to training the logistic regression classifier, within the same grouped 10-fold cross-validation structure described above, and compared using a paired Wilcoxon signed-rank test across folds.

3.8. Statistical software and reproducibility

All experiments were implemented in Python using TensorFlow/Keras (CNN training) and scikit-learn (classical classifiers, cross-validation utilities, and evaluation metrics). Random seeds were fixed (NumPy, Python's `random`, and TensorFlow) for reproducibility; however, minor residual run-to-run variation was observed (differences of at most one prediction out of 508 across repeated runs with identical seeds), attributed to GPU-level non-determinism in certain TensorFlow operations rather than to the fixed random seeds themselves.

4. Results

4.1. Pooled cross-validated performance across classifiers

Table 3 summarizes accuracy, precision (for both ADHD and Control classes), sensitivity, specificity, F1-score, and AUC for all six classifiers, pooled across 508 out-of-fold predictions from grouped 10-fold cross-validation. Pooled accuracy ranged from 64.57% (RF) to 70.28% (GNB), with AUC ranging from 0.711 (GNB) to 0.789 (CNN). Confusion matrices for all six classifiers are shown in Figure 3, and the corresponding ROC curves in Figure 4.

Notably, GNB and KNN – the two classifiers whose scikit-learn implementations do not support class-weight balancing – showed the largest sensitivity–specificity imbalance (GNB: 0.889 vs. 0.459; KNN: 0.785 vs. 0.514), consistent with a bias toward predicting the majority-leaning ADHD class (57% of epochs) rather than genuinely superior discrimination. This is visible directly in the confusion matrices (Figure 3), where GNB and KNN show substantially higher false-positive counts (119 and 107, respectively) than the class-weighted classifiers (53–105). By contrast, CNN – trained with a standard binary cross-entropy loss rather than explicit class weighting – showed the most balanced sensitivity–specificity profile (0.569 vs. 0.759) and the highest AUC (0.788) among all six classifiers, despite having the lowest raw sensitivity.

The AUC values in Table 3 and the confusion-matrix counts in Figure 3 were computed from an independent training run of the same pipeline as Figure 4, using an identical random seed. The two runs' AUC values differ by at

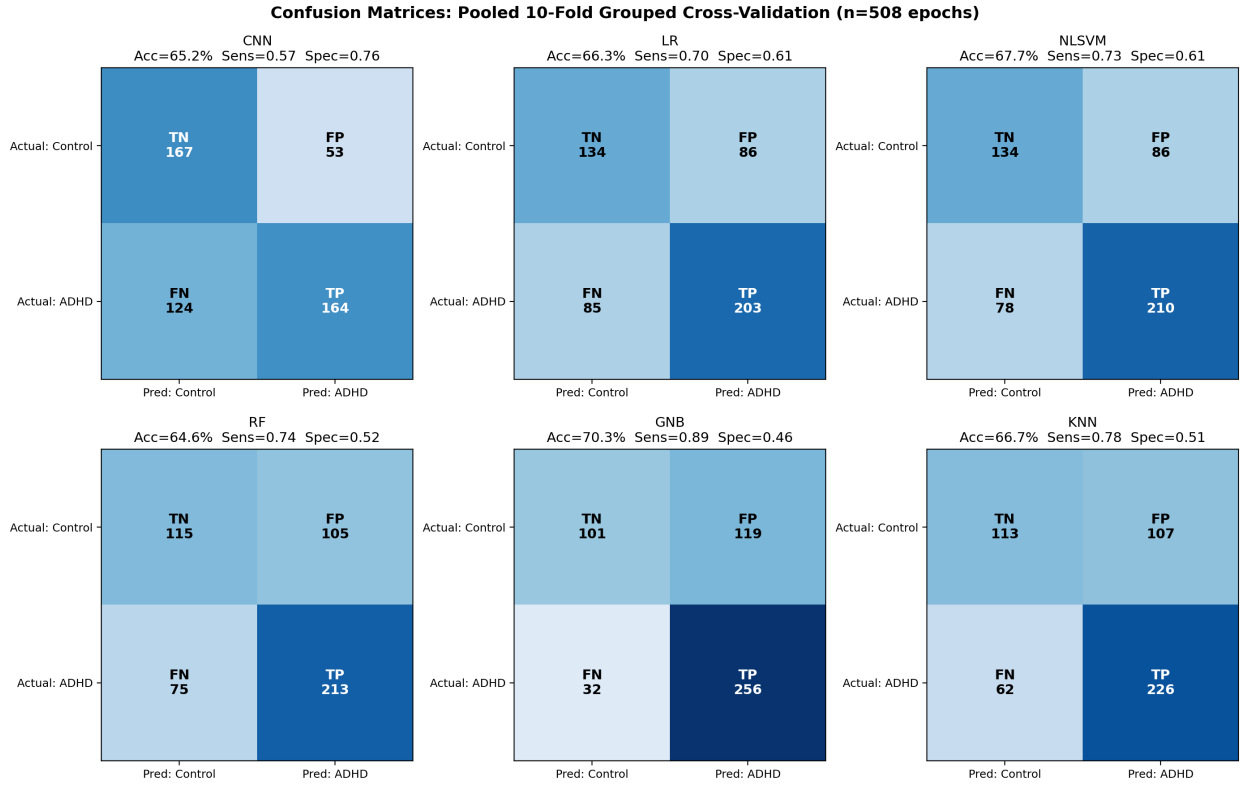


Figure 3: Confusion matrices for all six classifiers, pooled across 10 grouped cross-validation folds.

Table 4

Paired Wilcoxon signed-rank test results, CNN vs. each classical hybrid classifier (n=10 folds).

Comparison	<i>p</i> -value (accuracy)	<i>p</i> -value (AUC)
CNN vs. LR	0.9766	0.5703
CNN vs. NLSVM	0.6523	0.6523
CNN vs. RF	0.9102	0.2500
CNN vs. GNB	0.2031	0.1934
CNN vs. KNN	0.7344	0.4922

most 0.014 (e.g., CNN: 0.788 vs. 0.789), consistent with the minor GPU-level non-determinism described in Methods, and do not change any qualitative conclusion.

4.2. Per-fold variance across classifiers

Figure 5 shows per-fold accuracy for all six classifiers across the 10 grouped folds. Fold-to-fold variance was substantial for every classifier: standard deviations ranged from 13.6 percentage points (GNB) to 21.3 percentage points (CNN), with the best-performing fold for each classifier exceeding 88% accuracy and the worst falling below 50% (chance level) for CNN, LR, NLSVM, and RF. This indicates that a single train–test split – as used in the original single-split evaluation of this architecture – could, by chance, land on a particularly favorable or unfavorable subject grouping, yielding accuracy estimates that do not reflect a stable, generalizable measure of performance.

4.3. Statistical comparison between classifiers

Table 4 summarizes paired Wilcoxon signed-rank test results comparing the CNN’s per-fold accuracy and AUC against each of the five classical hybrid classifiers. No comparison reached statistical significance at $\alpha = 0.05$; the

ROC Curves: Pooled 10-Fold Grouped Cross-Validation
(n=508 out-of-fold predictions)

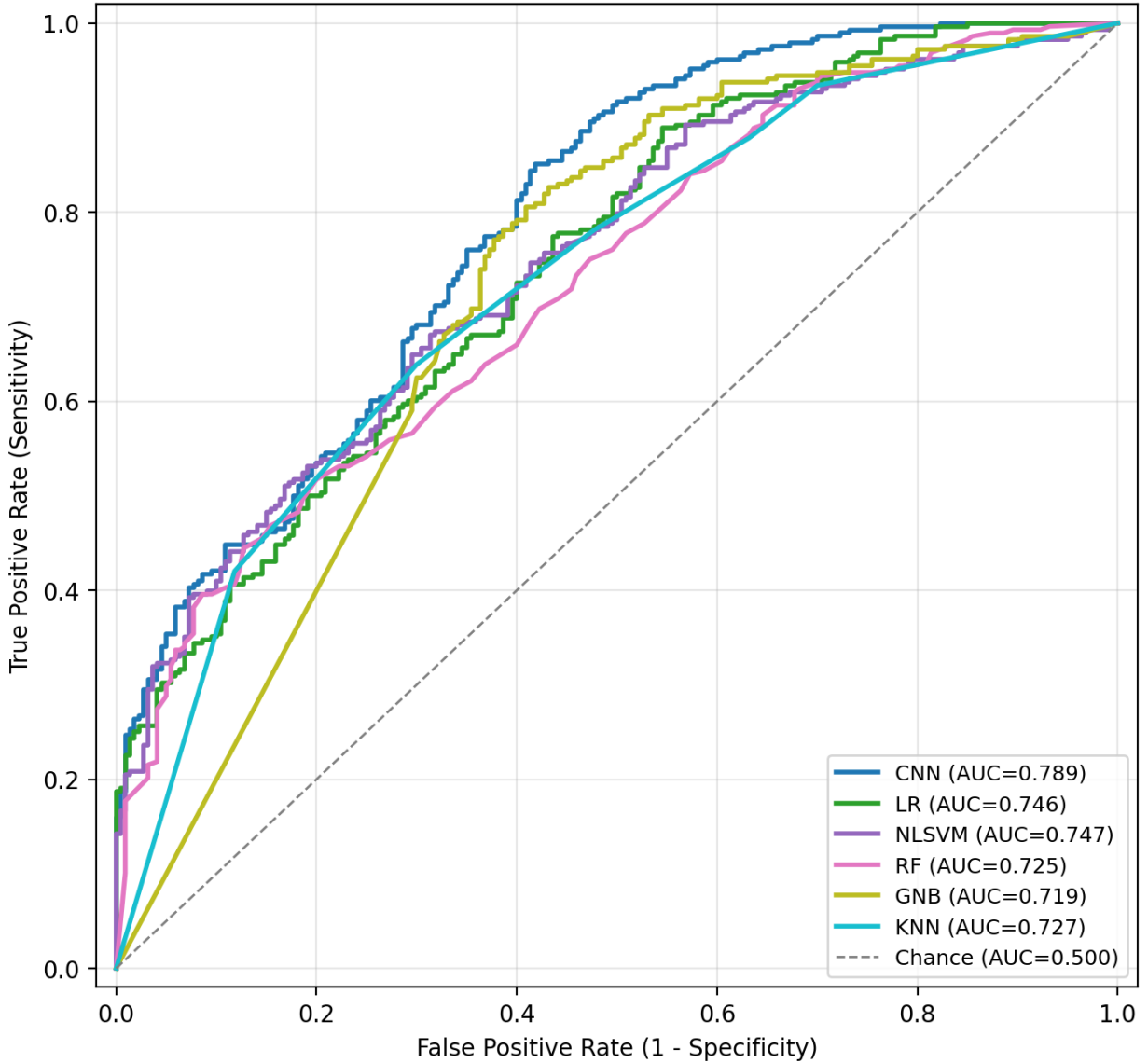


Figure 4: ROC curves for all six classifiers, pooled across 10 grouped cross-validation folds.

smallest p -value observed was 0.1934 (CNN vs. GNB, AUC). This contrasts with the substantially larger accuracy gap reported for the same comparison under single-split evaluation (see Section 4.4).

With only 10 paired observations, this test has limited statistical power; a non-significant result here should be interpreted as "not established with this sample size" rather than definitive evidence of equivalence between classifiers.

4.4. Comparison against previously reported single-split accuracy

Table 5 compares each classifier's pooled 10-fold cross-validated accuracy (this study) against the single-split test-set accuracy reported by TaghiBeyglou et al. [19] (Table 7 of that work, using Dense-1-layer CNN features). The gap between single-split and subject-independent cross-validated accuracy ranged from 11.00 percentage points (CNN) to

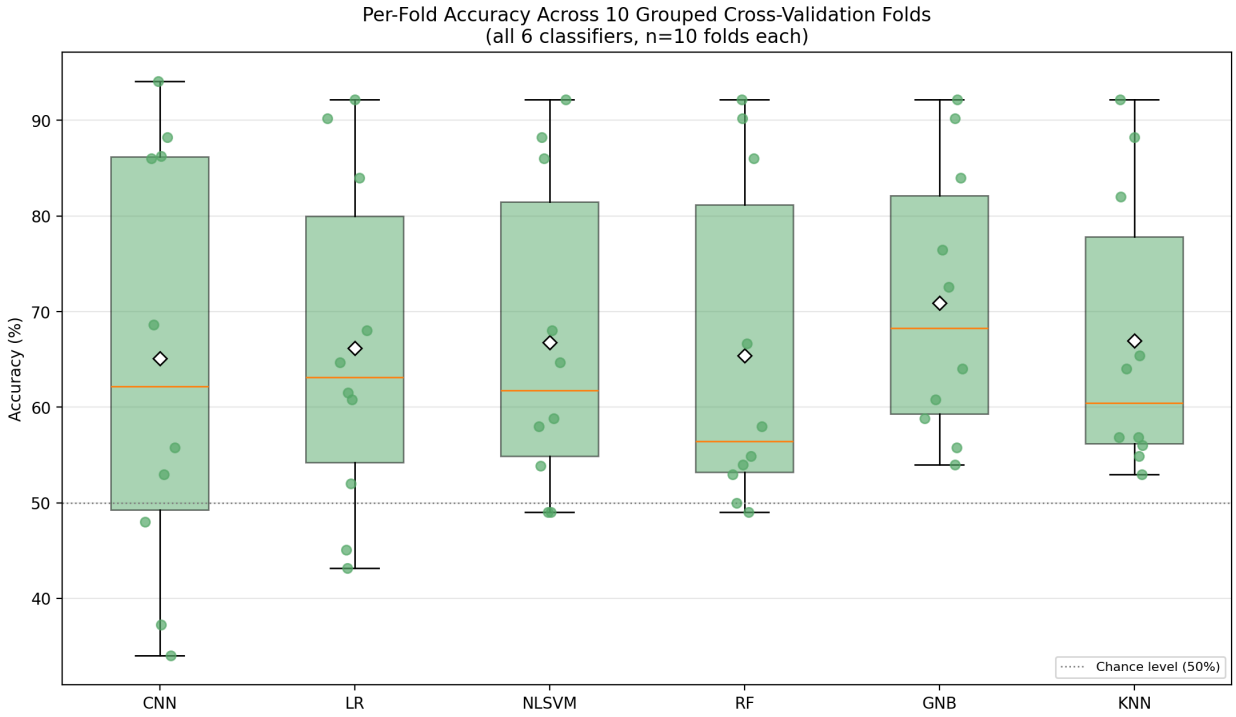


Figure 5: Per-fold accuracy across 10 grouped cross-validation folds for all six classifiers.

Table 5

Single-split accuracy (as reported in [19], Table 7) versus this study's pooled 10-fold cross-validated accuracy.

Classifier	Single-split (%)	10-fold pooled (%)	Gap (pts)
NLSVM	88.89	67.72	-21.17
LR	95.83	66.34	-29.49
RF	88.89	64.57	-24.32
GNB	87.50	70.28	-17.22
KNN	83.33	66.73	-16.60
CNN	76.16	65.16	-11.00

29.49 percentage points (LR) – notably, the largest gap corresponds to the classifier the original study identified as its best-performing method.

This comparison spans two different preprocessing and evaluation pipelines entirely (the original study's 512 Hz, 15,360-sample epochs versus our 128 Hz, 3,840-sample epochs, in addition to the difference in validation scheme), so this gap cannot be attributed solely to validation methodology. To help isolate the contribution of validation methodology specifically, we additionally compared our own pipeline's single-split accuracy against its own 10-fold pooled accuracy (i.e., holding preprocessing constant); this internal comparison showed a gap in the same direction for most classifiers (our single-split run: CNN 34.62%, LR 60.58%, NLSVM 63.46%, RF 61.54%, GNB 66.35%, KNN 51.92%, all below their respective 10-fold pooled values), though the magnitude and even direction of this gap varied considerably depending on which random split was drawn, as detailed in the following section.

4.5. Repeated single-split resampling

To assess whether previously reported single-split accuracies could be explained by favorable sample splitting alone, we performed 50 independent random 80/20 grouped splits using our own pipeline, training and evaluating all six classifiers on each split. Figure 6 shows the resulting distribution of accuracies per classifier, with each classifier's single-split accuracy as reported in [19] marked as a reference line.

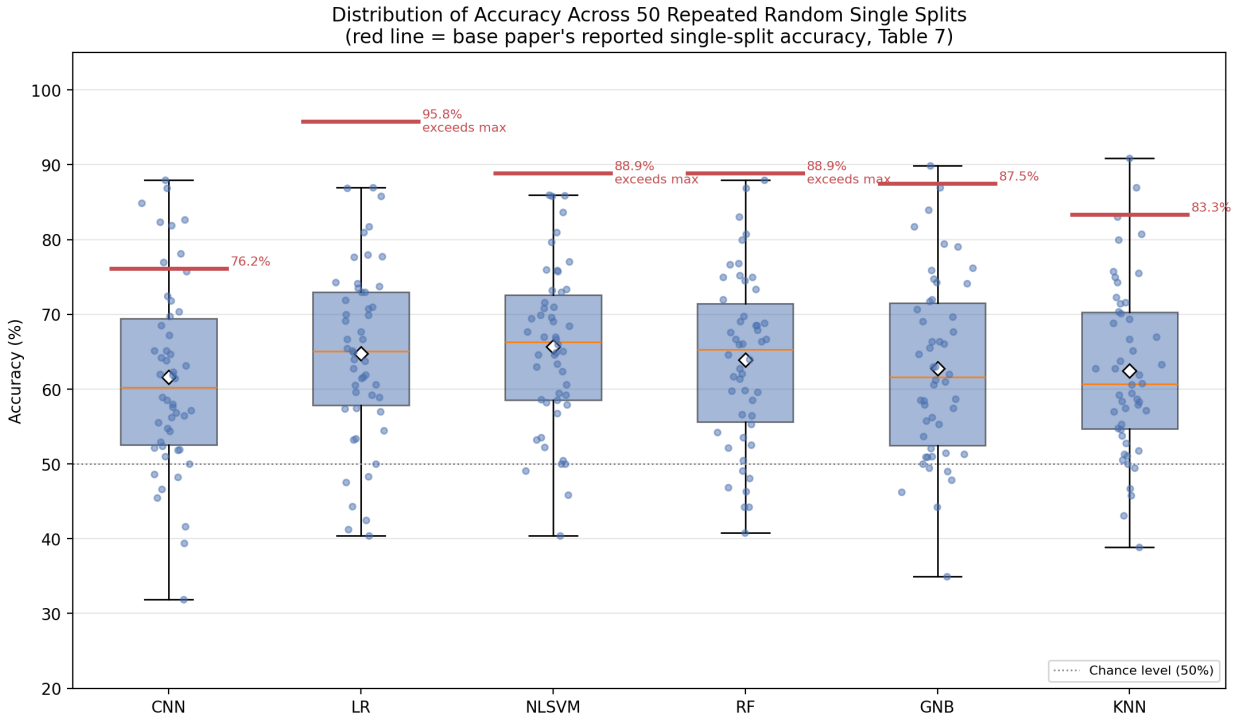


Figure 6: Distribution of accuracy across 50 repeated random single splits, per classifier, with previously reported single-split accuracy marked (red line).

For three classifiers – LR, NLSVM, and RF – the previously reported single-split accuracy (95.83%, 88.89%, and 88.89%, respectively) exceeded the maximum accuracy observed across all 50 repeated splits (87.00%, 86.00%, and 88.00%, respectively), placing these reported values at the 100th percentile of the resampled distribution. For the remaining three classifiers – CNN, GNB, and KNN – the previously reported accuracy fell within the range achievable by favorable sample splitting alone (84th, 98th, and 96th percentile, respectively). This suggests that favorable split selection alone is a plausible explanation for the previously reported performance of CNN, GNB, and KNN, but does not fully account for the reported performance of LR, NLSVM, and RF, for which some additional factor – potentially the higher temporal resolution of the original study's preprocessing pipeline (512 Hz vs. our 128 Hz), though this was not tested directly – may also contribute.

An important limitation of this analysis should be noted: because CNN weight initialization was not seeded independently across these 50 repeats (only the data split was fixed via a specified random state), this experiment conflates split-selection variance with CNN training-run variance. It should therefore be interpreted as a combined measure of "how favorable a split-and-training-run combination can be," rather than as an isolated measure of split variance alone.

4.6. Effect of CORAL domain adaptation

Figure 7 shows per-fold accuracy for logistic regression with and without CORAL-based domain adaptation [23], applied to the 64-dimensional Dense-1 features prior to classifier training. Mean accuracy across 10 folds was 66.16% (SD=16.74) without CORAL and 64.38% (SD=15.93) with CORAL. A paired Wilcoxon signed-rank test found no statistically significant difference between the two conditions ($p = 0.2969$). Examining the per-fold differences directly, CORAL improved accuracy in 1 of 10 folds, showed no change in 3 of 10 folds, and reduced accuracy in 6 of 10 folds.

Figure 7: Per-fold accuracy of logistic regression with and without CORAL domain adaptation, and the corresponding per-fold change.

5. Discussion

The central finding of this study is that reported performance for a raw-EEG, CNN-based ADHD classification framework [19] is substantially reduced under subject-independent, grouped cross-validation relative to the single-split evaluation originally reported. Pooled accuracy across all six classifiers fell in a narrow range of approximately 65–70% (Table 3), compared to a previously reported range of 76–96% (Table 5). This pattern is consistent with broader concerns raised about subject-level data leakage in translational EEG deep learning research [22], which explicitly identified [19] among the studies surveyed as using a validation approach susceptible to this issue.

A particularly notable finding is that the classifier identified as the best-performing method in the original study – logistic regression trained on CNN-derived features, reported at 95.83% single-split accuracy – showed both the largest single-split-to-cross-validation gap (29.49 percentage points, Table 5) and a reported accuracy that exceeded the maximum observed across 50 repeated random splits of our own pipeline (Figure 6). This suggests that the original study’s headline result may be particularly sensitive to the specific train–test partition used, an interpretation consistent with the high per-fold variance observed for this and other classifiers (Figure 5, standard deviations of 13.6–21.3 percentage points). At the same time, the original study’s exact preprocessing pipeline (512 Hz sampling, 15,360-sample epochs) was not tested directly, so this difference in temporal resolution cannot be ruled out as a contributing factor to the gap for LR, NLSVM, and RF specifically; this remains an open question for future work, and this attribution of the gap primarily to validation methodology should be treated as a plausible interpretation rather than a fully isolated causal claim.

Under subject-independent evaluation, no classifier – classical or CNN-based – demonstrated a statistically significant advantage over the others (Table 4, all $p > 0.05$). This is a materially different conclusion from the original single-split comparison, and suggests that claims of hybrid-classifier superiority in raw-EEG ADHD classification should be evaluated cautiously absent subject-independent validation. Classifiers without built-in class-weight balancing (GNB, KNN) showed pronounced sensitivity–specificity imbalance, achieving comparatively high accuracy and F1-scores that appear to be driven substantially by a bias toward predicting the majority-leaning class rather than by superior discrimination (Table 3, Figure 3). This is a practical reminder that aggregate accuracy or F1-score alone can be misleading under class imbalance, and that sensitivity and specificity should be reported and examined separately, particularly for a clinical screening application.

CORAL-based domain adaptation, which has shown benefit for cross-subject and cross-dataset generalization in other EEG classification tasks such as emotion recognition, did not yield a statistically significant improvement in this setting ($p = 0.2969$), and, if anything, showed a directionally negative effect across most folds (Figure 7). This may plausibly reflect the relatively small covariance mismatch already present between training and test fold features in this dataset, leaving limited room for a covariance-alignment technique to provide benefit, though this explanation was not directly tested and should be treated as speculative; it is also possible that the small per-fold sample size (approximately 50 test epochs per fold) limits the reliability of the target-domain covariance estimate CORAL depends on. To our knowledge, CORAL has not previously been evaluated specifically for cross-subject generalization in raw-EEG, CNN-based ADHD classification, and this negative result should itself be considered a useful data point for future work considering domain adaptation approaches for this task.

5.1. Limitations

Several limitations should be considered when interpreting these results. First, the sampling rate of the dataset used in this study (assumed to be 128 Hz, based on the original dataset description in [19]) could not be independently confirmed, as no sampling-rate metadata was present in the released data files; this should be treated as an assumption rather than a verified fact. Second, the repeated single-split analysis (Section 3.5) conflated split-selection variance with CNN training-run variance due to unfixed random weight initialization across repeats, limiting the precision of the percentile-based comparisons reported. Third, statistical comparisons in this study (Wilcoxon signed-rank tests) were based on only 10 paired folds, limiting statistical power; non-significant results should not be interpreted as proof of equivalence between classifiers. Fourth, the original study’s exact 512 Hz preprocessing pipeline was not tested directly, so the contribution of validation methodology cannot be fully separated from that of preprocessing differences in explaining the observed performance gap. Finally, minor run-to-run numerical variation was observed (differences up to approximately 1 percentage point in pooled accuracy, and up to 0.014 in AUC) across repeated executions of the same pipeline with fixed random seeds, attributed to GPU-level non-determinism in certain TensorFlow operations;

this should be kept in mind when comparing exact figures across tables and figures in this paper, though it does not affect any qualitative conclusion drawn.

6. Conclusion

This study examined the extent to which reported classification performance for a raw-EEG, CNN-based ADHD detection framework [19] holds under subject-independent evaluation. Using grouped 10-fold cross-validation on a publicly available pediatric ADHD/Control EEG dataset, pooled accuracy across six classifiers (a CNN and five classical hybrid classifiers) fell in the range of approximately 65–70%, substantially below the 76–96% range previously reported using a single fixed train–test split. No classifier demonstrated a statistically significant advantage over the others under this stricter validation scheme, in contrast to the original single-split comparison, in which logistic regression substantially outperformed the CNN alone.

Repeated random single-split resampling further indicated that, for three of the six classifiers evaluated (logistic regression, non-linear support vector machine, and random forest), previously reported single-split accuracy exceeded the maximum achievable through favorable sample splitting alone across 50 repeated trials, suggesting that factors beyond split selection – potentially including differences in data resolution between the original and present study – may also contribute to this gap. For the remaining three classifiers (CNN, Gaussian naive Bayes, and k -nearest neighbors), previously reported accuracy fell within the range attributable to favorable splitting alone.

CORAL-based unsupervised domain adaptation, despite demonstrated benefit in other cross-subject EEG classification tasks, did not produce a statistically significant improvement in cross-subject generalization for this task. This negative result is reported as a contribution in its own right, given the absence, to the authors' knowledge, of prior evaluations of domain adaptation for raw-EEG, CNN-based ADHD classification specifically.

Taken together, these findings support growing concerns in the translational EEG deep learning literature regarding the reliability of single-split performance estimates, and underscore the importance of subject-independent validation when assessing the clinical potential of EEG-based biomarkers for ADHD. Future work should directly evaluate whether the original study's higher-resolution preprocessing pipeline narrows the observed performance gap, and should consider alternative domain adaptation or regularization strategies, together with larger and more diverse datasets, to further improve subject-independent generalization in this task.

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CRedit authorship contribution statement

Abinaya G: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Declaration of Competing Interest

The author declare that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical statement

This study used only publicly available datasets. No new human participants were recruited, and no new data were collected. All ethical approvals for the original data collection were obtained by the respective dataset authors as described in their primary publications. No additional institutional ethics review was required for this secondary analysis of open data.

Data Availability

The EEG dataset used in this study is publicly available via IEEE DataPort: A. Motie Nasrabadi, A. Allahverdy, M. Samavati, M.R. Mohammadi, "EEG data for ADHD/Control children," IEEE DataPort, 2020, <https://doi.org/10.>

21227/rzfh-zn36. The sampling rate of the released data could not be independently confirmed from the provided files and was assumed to be 128 Hz based on the description of the original dataset in [19]; this assumption is stated explicitly in the Methods section and should be verified independently by future users of this dataset. Code used for preprocessing, model implementation, and analysis is available from the corresponding author upon reasonable request.

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Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ethics Statement

This study used only publicly available, de-identified EEG data. No new human participants were recruited, and no new data were collected by the author. The dataset used (Nasrabadi et al., "EEG data for ADHD/Control children," IEEE DataPort, 2020, <https://doi.org/10.21227/rzfh-zn36>) was collected and released by its original authors; all ethical approvals and participant consent procedures for the original data collection were the responsibility of those original authors, as described in their primary publications and dataset documentation. No additional institutional ethics review was required for this secondary, computational re-analysis of open, de-identified data.